SIMATS ENGINEERING

Regex for real-world text

A practical pattern-matching layer for NLP

TECHNICAL PRESENTATION

Regular Expressions for Real-World Text Processing

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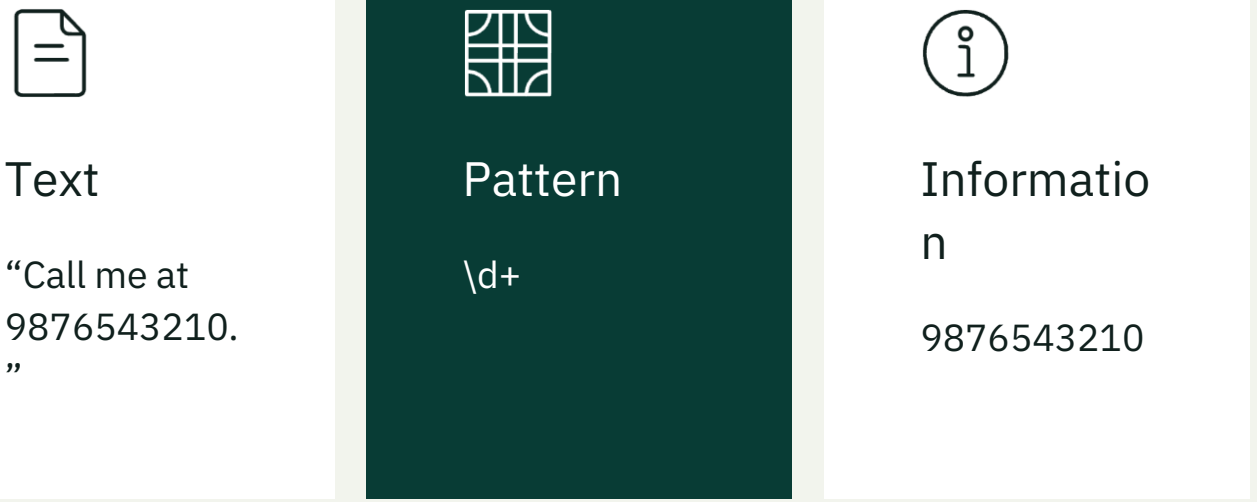
Course: DSA0311- Natural Language Processing

THE PRACTICAL LAYER

Regex gives NLP a practical pattern-matching layer

NLP helps computers work with human language. Regular expressions, or regex, describe text patterns that are useful for finding, extracting, validating, and cleaning text.

Example: finding a phone number inside a sentence.



BEGINNER TOOLKIT

Five symbols cover many beginner tasks

Symbol	Meaning	Typical use
\d	Any digit	Find numbers such as 7 or 2026
\w	Word character	Match letters, digits, or underscore
\s	Whitespace	Find spaces, tabs, or line breaks
.	Any character	Match one character in a position
+	One or more	Repeat the preceding pattern

A COMPLETE MICRO-EXAMPLE

One pattern can extract the number directly

01

Input text

“My contact number is 9876543210.”

02

Regex pattern matching

\d+ finds one or more digits.

03

Extracted number

9876543210

The pattern identifies the matching text without manually selecting it.

WHERE IT HELPS

Regex supports everyday NLP preprocessing



Email

Pattern detection



Phone

Number extraction



Cleaning

Remove unwanted text



Logs

Find recurring entries



Retrieval

Locate useful terms

USE THE RIGHT TOOL

Regex is fast, but it is not language understanding

Regex is excellent at matching visible patterns. Its limits appear when meaning depends on context, ambiguity, or the full sentence.

Advantages

- Fast pattern matching
- Reusable patterns
- Useful for text extraction and cleaning

Limitations

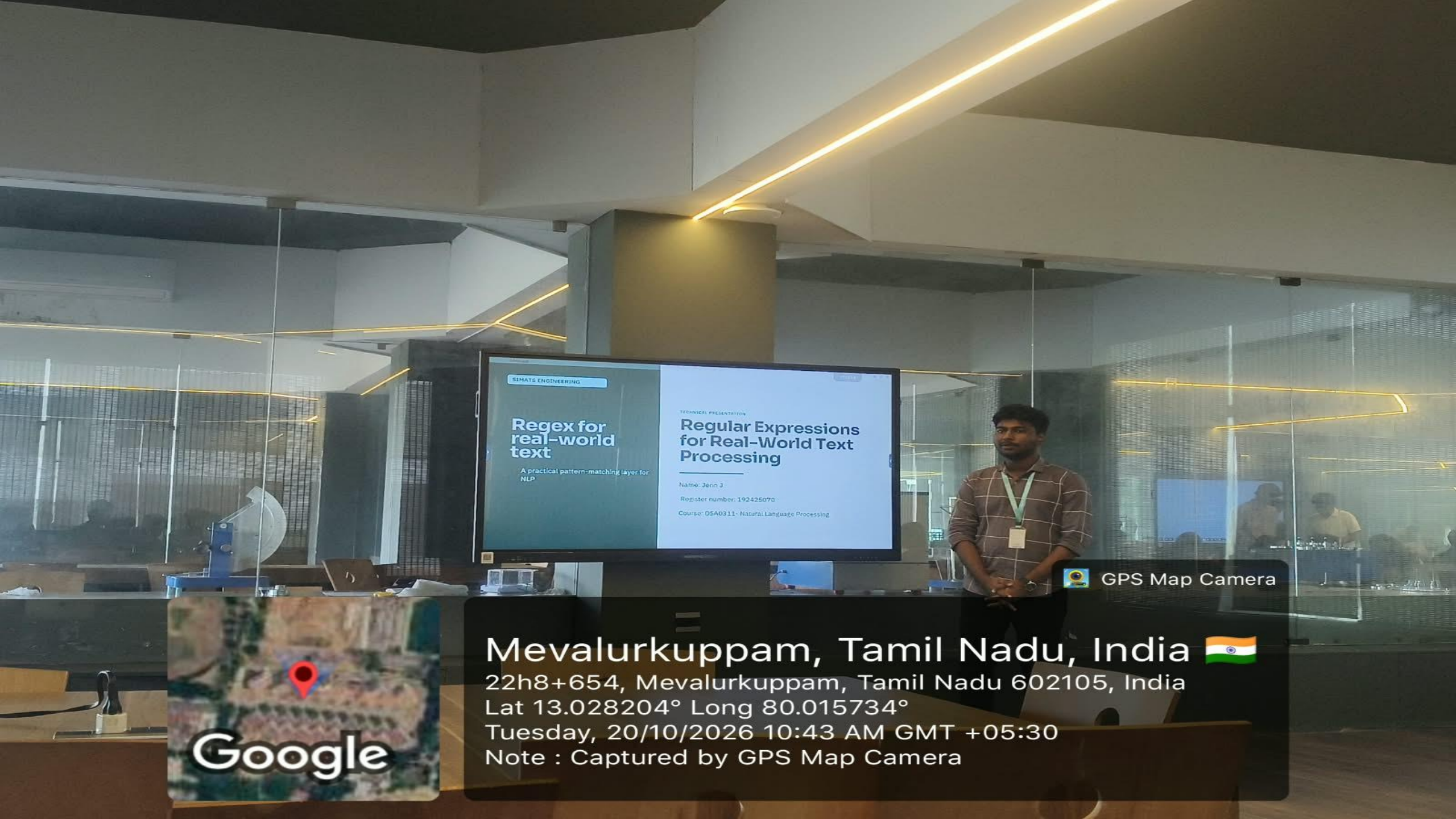
- Complex patterns can be difficult to maintain
- Patterns may incorrectly accept or reject text
- Regex cannot understand the full meaning of a sentence

THE TAKEAWAY

Regex finds patterns; NLP adds understanding

Regex is a useful tool for identifying text patterns. It helps extract information and clean text. It supports NLP preprocessing but does not replace language understanding.

Thank You


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